

Data Science Fundamentals

1st Week – 2nd session

2024/25

Python – What is it?

Object-oriented programming language (just like Java, C++, etc.)

Interpreted language (different than compiled languages – Java, C++)

Executes code line-by-line (also different than compiled languages)

Dynamically typed: During execution, the data type is checked and inferred automatically (quite different from Java, C, etc.)

Python – A bit of history

Created in the late 80s by Guido van Rossum

Named after the famous BBC television series – Monty Python.

It was the successor of the ABC programming language, as it was able to handle exceptions and was integrated into the Amoeba operating system.

Python – Why use it?



Python – Advantages & Disadvantages



Advantages

- Excellent for data exploration (and thus, for data science)
- A large collection of powerful libraries that are free to use and simple to install (pip/conda install package name).
- Very active scientific and academic community!
- Write Once, Run Everywhere!, just like Java

Disadvantages

- Slower (because it's interpreted line by line – but makes debugging much easier!)
- Sometimes, it is inefficient in memory consumption
- Weak in mobile application development

Python – A universe of applications



Python – Some Python libraries for Data Science



Python – Why Python for Data Science instead of R?

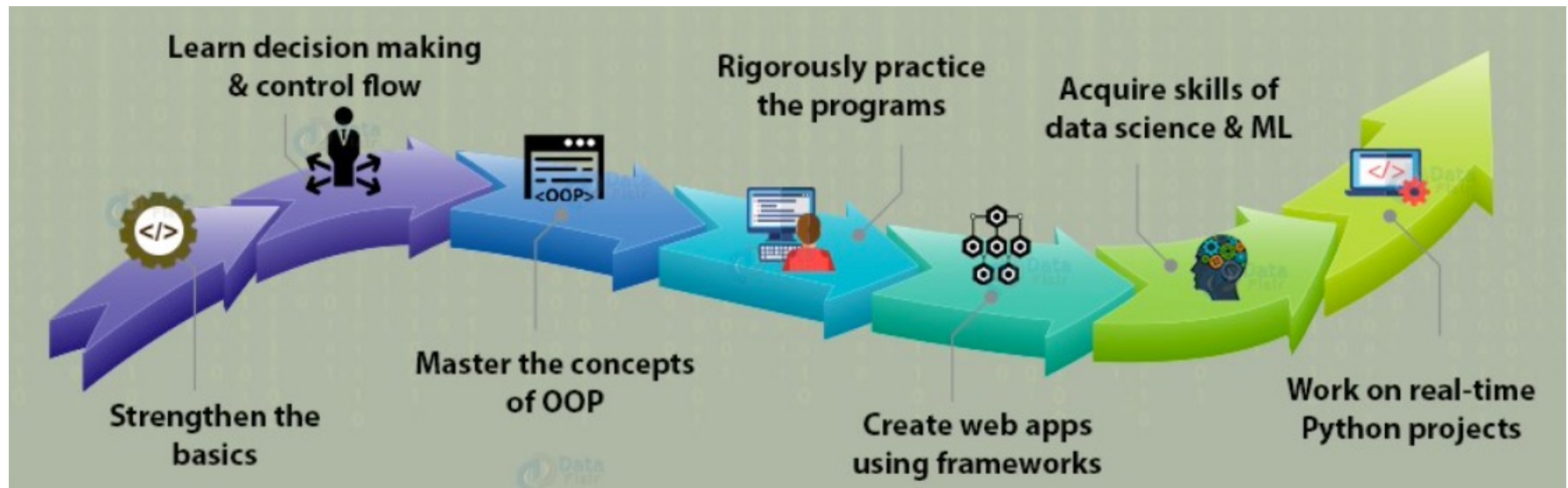
Parameter	R	Python
Objective	Data Analysis and Statistical Modeling	Data Science, Web Development, Embedded Systems
Workability	Consists of many easy to use packages	Can easily perform matrix computation as well as optimization
Integration	Locally Run Programs	Programs integrated with web-app for easy deployment
Database Handling Capacity	Poses problem for handling large dataset	Can handle large data easily without any fault
IDE	Rstudio, R GUI	Spyder, IPython, Jupyter Notebook
Essential Packages and library	ggplot2, tidyverse, caret	Numpy, pandas, scipy, scikit-learn, TensorFlow

Comparison between R Programming and Python



The image shows the R logo (a blue 'R' inside a white circle) on the left, a red 'VS' with a lightning bolt in the center, and the Python logo (two interlocking snakes, one blue and one yellow) on the right.

Python – Learning flow in Data Science

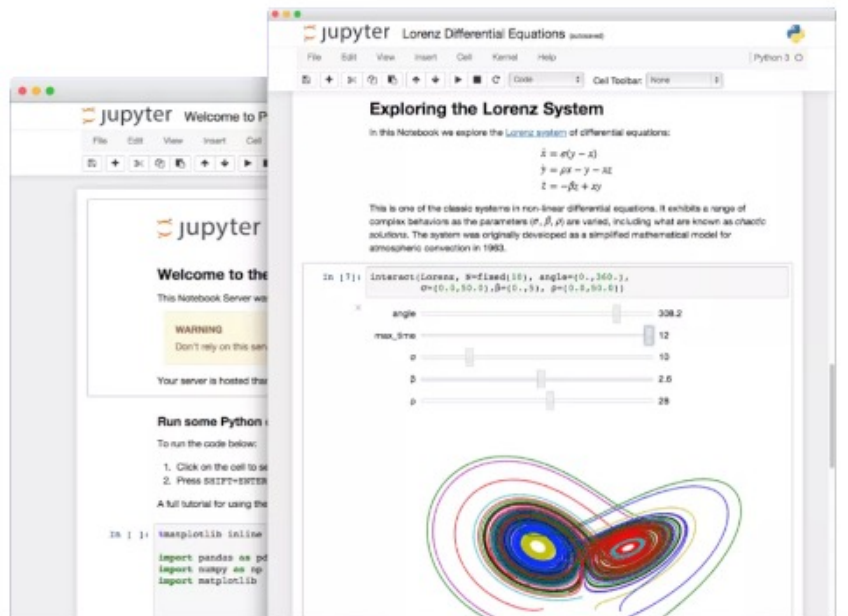


JupyterLab – What is it?

- Inspired by the scientist's notebook
- Open-source platform for exploratory and interactive computing
- IPython Notebook 2011
- Jupyter Notebook 2014
- Supports a variety of programming languages
- Jupyter is an acronym for Julia (Ju), Python (Py), and R
- JupyterLab 2018, 2019, 2021–22



<https://jupyter.org/>



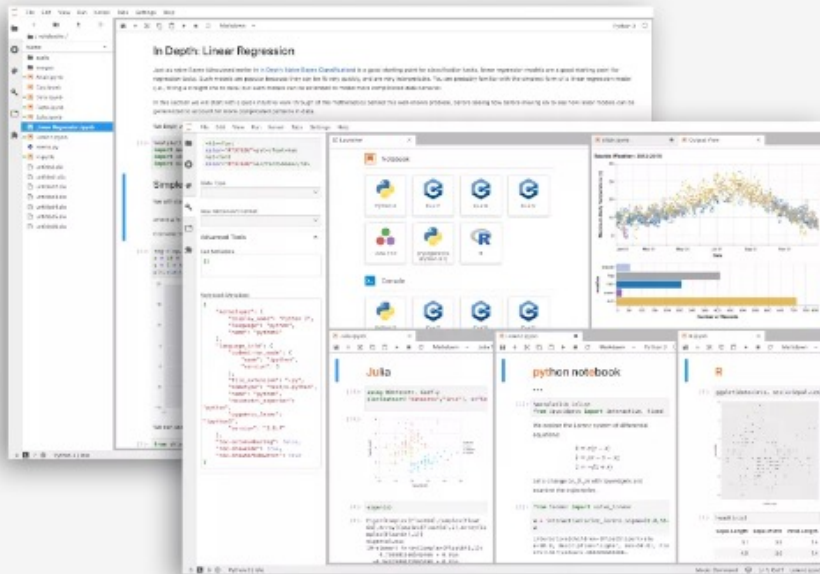
Jupyter Notebook: The Classic Notebook Interface

The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

Try it in your browser

Install the Notebook

<https://jupyter.org/>



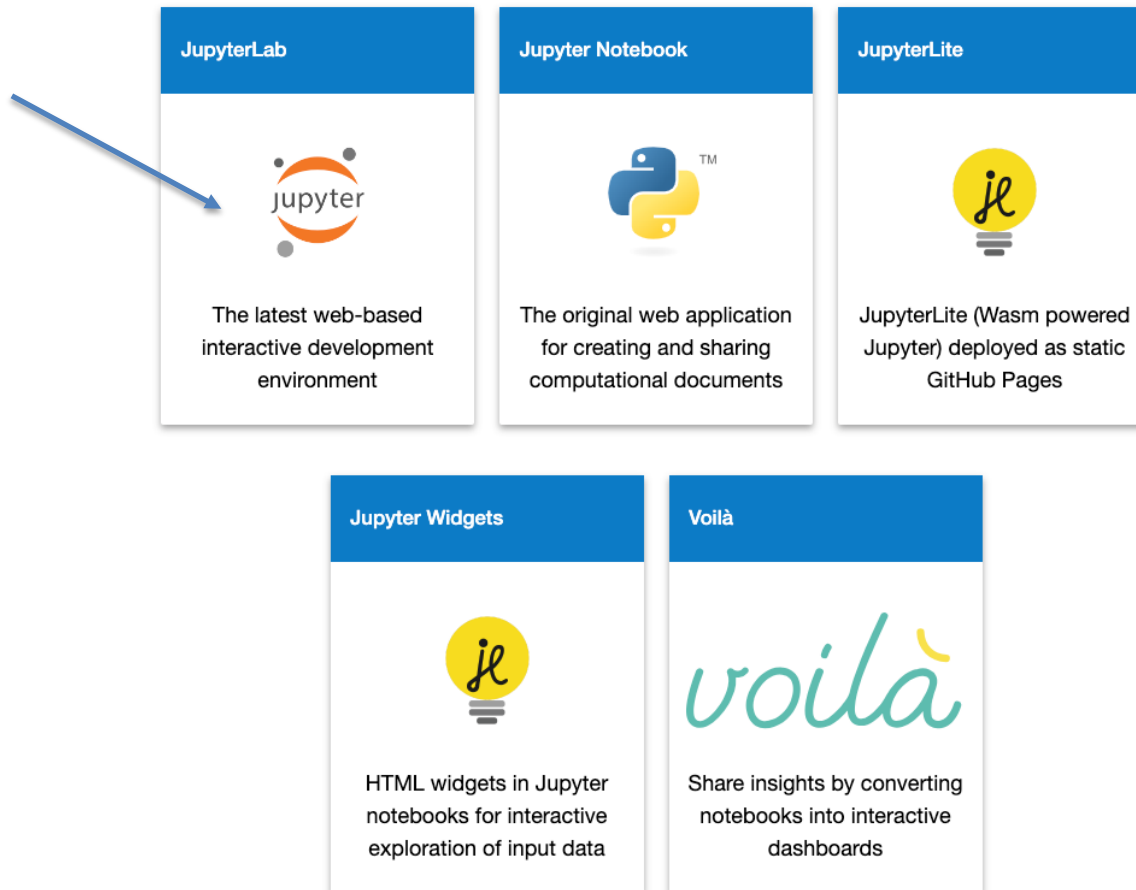
JupyterLab: A Next-Generation Notebook Interface

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning. A modular design invites extensions to expand and enrich functionality.

Try it in your browser

Install JupyterLab

Let's try in our browser...



JupyterLab Welcome Tour

Help -> Welcome Tour

JupyterLab documentation

<https://jupyterlab.readthedocs.io/en/stable/>

From JupyterLab documentation

- A computational notebook is a shareable document that combines computer code, plain language descriptions, data, rich visualizations like 3D models, charts, graphs and figures, and interactive controls.
- A notebook, along with an editor like JupyterLab, provides a fast interactive environment for prototyping and explaining code, exploring and visualizing data, and sharing ideas with others.

JupyterLab installation

- JupyterLab can:
 - run online – <https://jupyter.org/try-jupyter/lab/index.html>
or
 - can be installed and run in your computer
 - 1st install Python (for using pip)
 - then execute in a terminal: *pip install jupyterlab*
 - and then run in a terminal: *jupyter lab*

both versions will run on your browser

- Supported browsers:
 - Firefox
 - Chrome
 - Safari

How to Use JupyterLab

an introduction to JupyterLab, Markdown and Python on JupyterLab, on YouTube

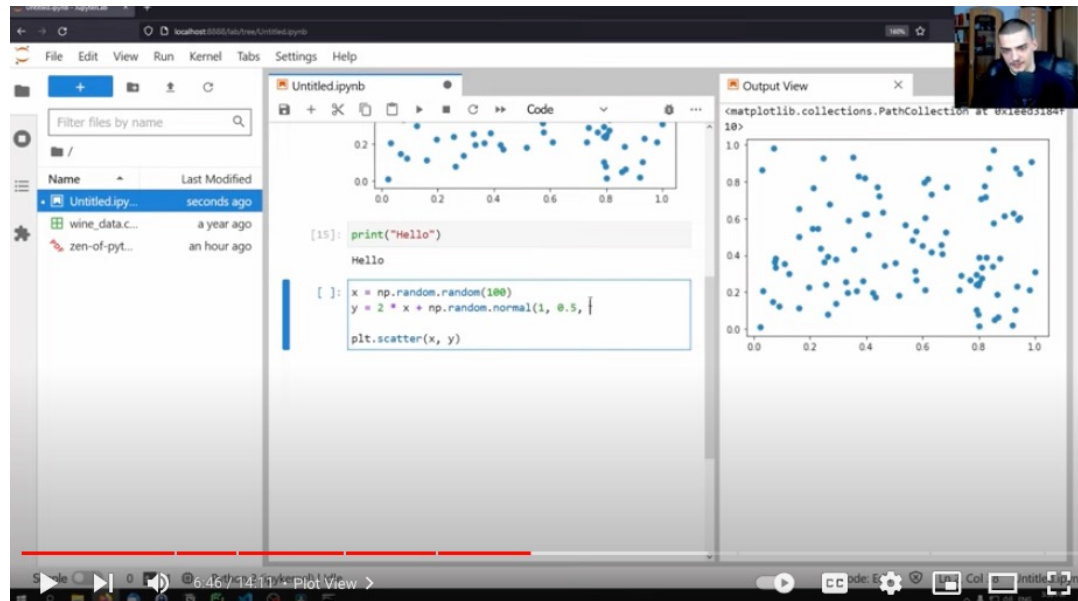
<https://www.youtube.com/watch?v=A5YyoCKxEOU&t=294s>



Jupyter Lab is AWESOME For Data Science

also on YouTube, more advanced on how to use Python on notebooks

<https://www.youtube.com/watch?v=7wf1HhYQiDg>



The JupyterLab Interface

Menu Bar

Left and Right Sidebar

The screenshot displays the JupyterLab interface with the following components:

- Menu Bar:** Located at the top, containing options: File, Edit, View, Run, Kernel, Tabs, Settings, Help.
- Left Sidebar:** Contains a file browser showing a directory structure under `/data/`. Files listed include `bar.vl.json`, `Dockerfile`, `iris.csv`, `japan_me...`, `Museums...`, `README...`, and `zika_asse...`. All files are marked as "yesterday".
- Main Editor:** Displays a Jupyter notebook titled `Data.ipynb`. The current cell contains the following code:

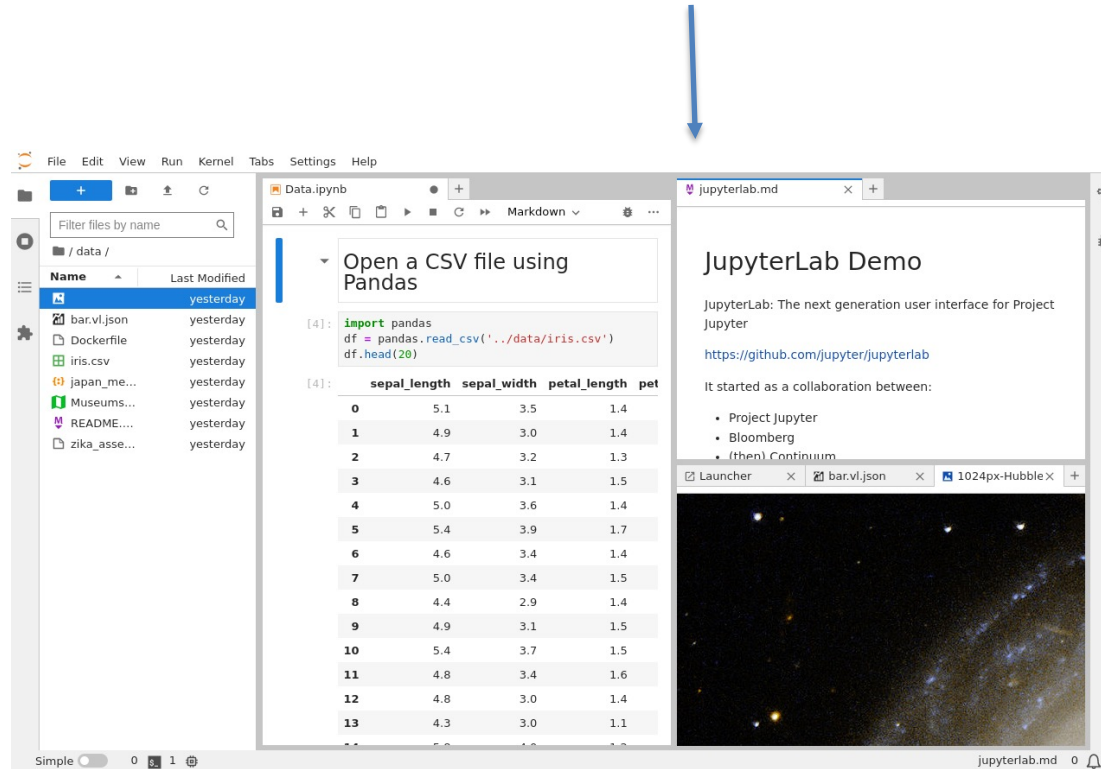
```
[4]: import pandas
df = pandas.read_csv('../data/iris.csv')
df.head(20)
```

The output shows the first 20 rows of the Iris dataset, with columns: `sepal_length`, `sepal_width`, `petal_length`, and `pet`.
- Right Sidebar:** Contains a markdown viewer for `jupyterlab.md`. The content includes:
 - JupyterLab Demo**
 - Description: "JupyterLab: The next generation user interface for Project Jupyter"
 - Link: <https://github.com/jupyter/jupyterlab>
 - Text: "It started as a collaboration between:"
 - List:
 - Project Jupyter
 - Bloomberg
 - (then) Continuum

At the bottom of the right sidebar, there is a tab bar showing `Launcher`, `bar.vl.json`, and `1024px-Hubble`. Below the tabs is a large image of a galaxy.

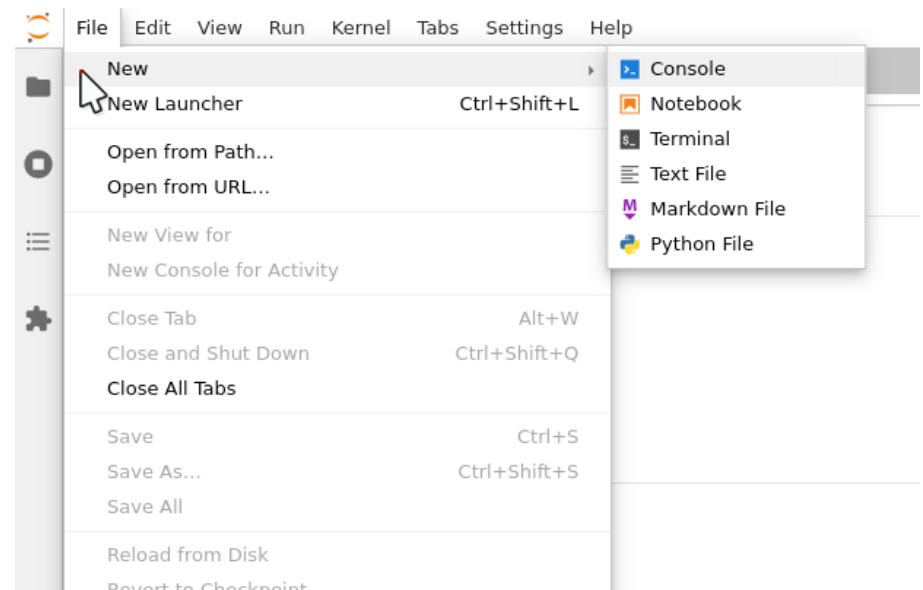
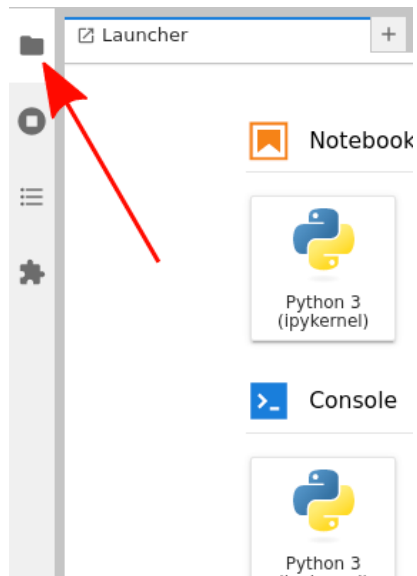
The JupyterLab Interface

Main Work Area



Context Menus can be accessed by right-clicking on each element

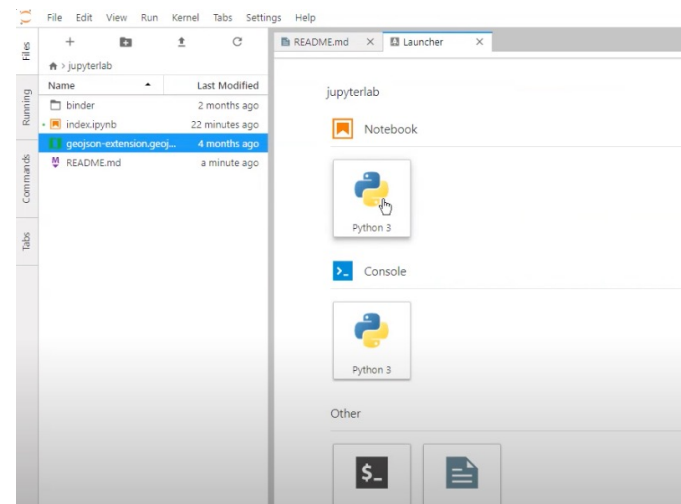
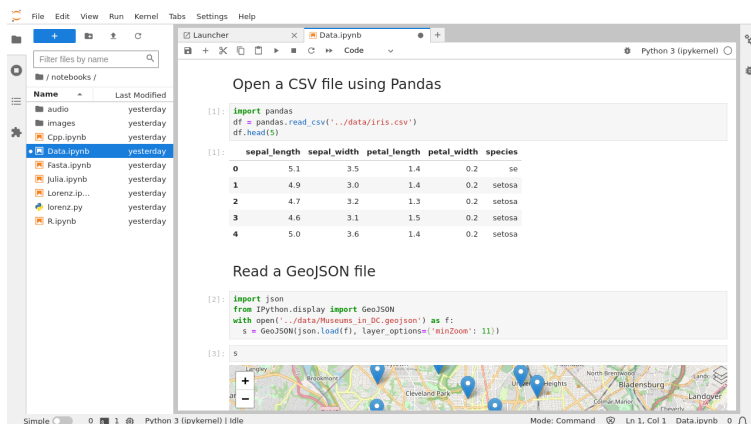
Creating/opening/managing Files



also, you can drag-and-drop files to the main work area

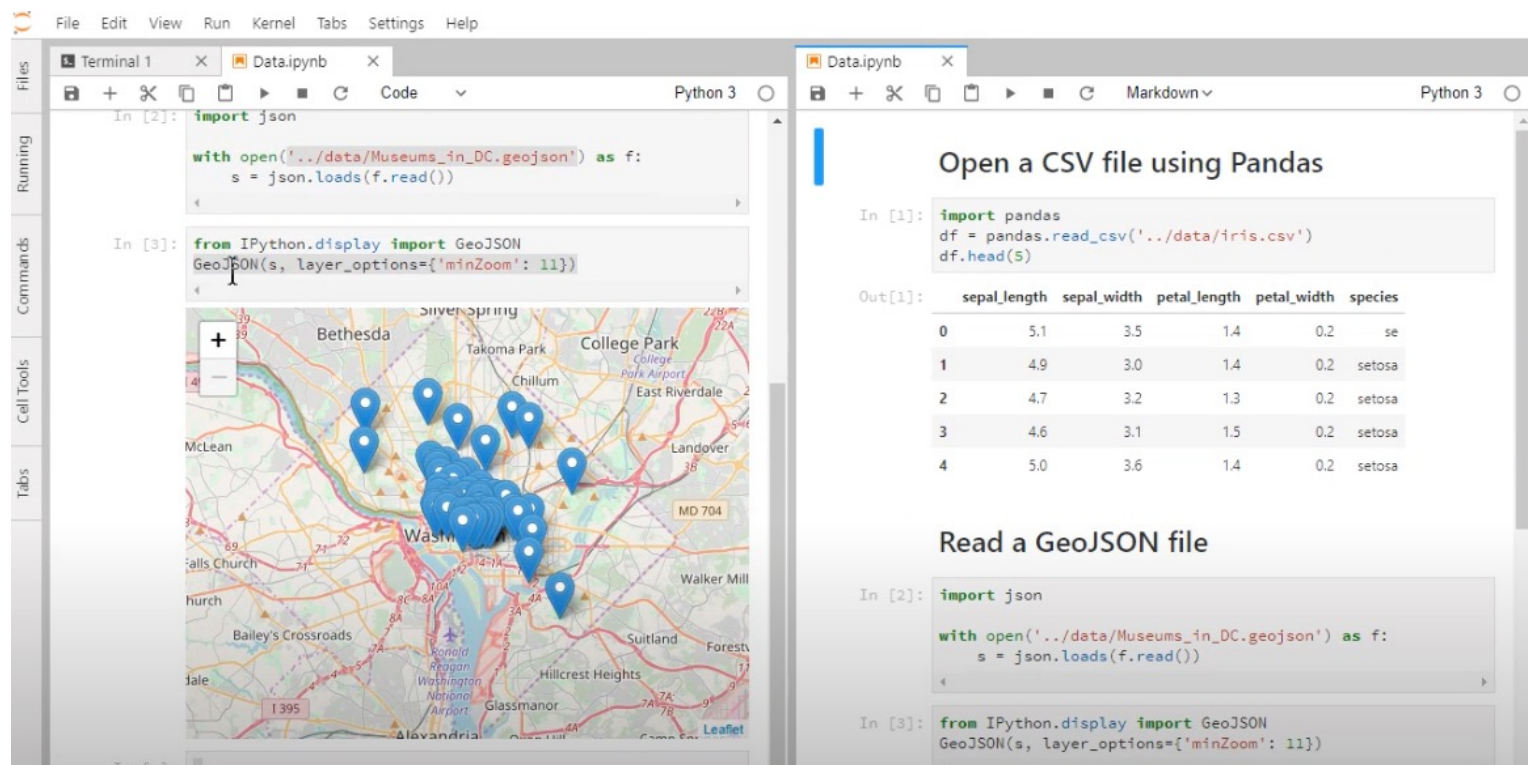
Notebooks

Jupyter notebooks are documents that combine live runnable code with narrative text (Markdown), equations (LaTeX), images, interactive visualizations and other rich output



Create a notebook by clicking the + button in the file browser and then selecting the Python kernel in the new Launcher tab

Create multiple synchronized views of a single notebook



The image displays two side-by-side JupyterLab notebooks, illustrating how a single notebook can be viewed in multiple synchronized instances. Both notebooks are running on Python 3.

Left Notebook (Data.ipynb):

- Code:**

```
In [2]: import json

with open('../data/Museums_in_DC.geojson') as f:
    s = json.loads(f.read())

In [3]: from IPython.display import GeoJSON
GeoJSON(s, layer_options={'minZoom': 11})
```
- Output:** A map of Washington, D.C. showing the locations of museums marked with blue pins. The map includes labels for various areas like Bethesda, Silver Spring, College Park, Takoma Park, and Washington.

Right Notebook (Data.ipynb):

- Code:**

```
In [1]: import pandas
df = pandas.read_csv('../data/iris.csv')
df.head(5)
```
- Output:** A table showing the first 5 rows of the Iris dataset.

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	se
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Code:

```
In [2]: import json

with open('../data/Museums_in_DC.geojson') as f:
    s = json.loads(f.read())

In [3]: from IPython.display import GeoJSON
GeoJSON(s, layer_options={'minZoom': 11})
```


Markdown in a JupyterLab notebook

Search for "Markdown in Jupyter Notebook Tutorial"

Markdown in a JupyterLab notebook

Headings

(Header 1, title)

(Header 2, major headings)

(Header 3, subheadings)

(Header 4)

(Header 5)

(Header 6)

<h1>Header 1,title<h1>

<h2>Header 2,major
headings<h2>

<h3>Header
3,subheadings<h3>

<h4>Header 4<h4>

<h5>Header 5<h5>

<h6>Header 6<h6>

Markdown in a JupyterLab notebook

Blockquotes

>This is good

<blockquote>This is good</blockquote>

Markdown in a JupyterLab notebook

Code Section

Using Markdown

```
```Python  
str = "This is block level code"
print(str)
```
```

Using Markup Tags

```
<code>Python  
str = "This is a block level  
code"  
print(str)  
</code>
```

Markdown in a JupyterLab notebook

Line Break

`<br>`

Bold Text

`<b>This is bold text </b>`

`** This is bold text`

`__ This is bold text`

Markdown in a JupyterLab notebook

Italic text

`<i>This is italic text </i>`

`* This is italic text`

`_ This is italic text`

Markdown in a JupyterLab notebook

Horizontal Line

Ordered List

`ol>`

`<li>Fish</li>`

`<li>Eggs</li>`

`<li>Cheese</li>`

`</ol>`

1. Fish

2. Eggs

3. Cheese

Markdown in a JupyterLab notebook

Unordered List

```
<ul>  
<li>Fish</li>  
<li>Eggs</li>  
<li>Cheese</li>  
</ul>
```

- Fish
- Eggs
- Cheese

Markdown in a JupyterLab notebook

Internal Link

Define a section with an ID:

[Section title](#division_ID)

Then you can make an internal link:

go to Section title

The result would be the link

[go to Section title](#)

Markdown in a JupyterLab notebook

External Link

```
<a href="https://www.google.com" >Link to  
Google</a>
```

Markdown in a JupyterLab notebook

Table

```
<table style="width:20%">  
  <tr>  
    <th><Name></th>  
    <th><Address></th>  
    <th><Salary></th>  
  </tr>  
  <tr>  
    <td><Hanna></td>  
    <td><Brisbane></td>  
    <td><4000></td>  
  </tr>  
</table>
```

Markdown in a JupyterLab notebook

Images

```

```